Section A – Network Communication and Switching Techniques

Question 1. Circuit Switching vs. Packet Switching in Industrial Communication

An industrial manufacturing facility requires communication between PLCs, HMIs, and supervisory control systems. Two communication approaches are being considered: circuit switching and packet switching. Compare these two approaches in terms of resource allocation, communication delays, network efficiency, and suitability for industrial automation. Explain why packet switching is widely used in modern industrial Ethernet networks.

Model Answer:

1. Circuit switching establishes a dedicated communication path between two devices before data transmission begins and reserves network resources for the entire connection.
2. Packet switching divides messages into smaller packets that share the available network infrastructure with traffic from other devices.
3. Circuit switching provides more predictable bandwidth and transmission delays because resources remain reserved throughout communication.
4. Packet switching uses network bandwidth more efficiently because communication resources are utilized when devices have data to transmit, although congestion can produce delays and jitter.
5. Packet switching is widely used in modern industrial Ethernet networks because it supports flexible, scalable, and cost-effective communication among PLCs, HMIs, computers, and servers.

Lecture reference: Section 2.1 – Circuit Switching and Packet Switching, Slides 3–6.

Question 2. Datagram vs. Virtual Circuit Packet Switching

A manufacturing company needs to transfer production information between several plants through a communication network. The network designer is evaluating datagram-based communication and virtual circuit packet switching. Analyze the operational differences between these two services and explain how packet routing, connection establishment, packet ordering, and network performance influence the selection of an appropriate communication method.

Model Answer:

1. Datagram communication is connectionless, meaning individual packets can be transmitted without establishing a predetermined communication path.
2. Each datagram is handled independently and may follow a different route, experience different delays, arrive out of sequence, or be lost.
3. Virtual circuit switching establishes a logical path before communication begins, and packets normally follow the same logical route.
4. Virtual circuit communication generally provides more predictable forwarding and packet ordering, although the communication resources remain shared.
5. Datagram services such as IP are flexible and scalable, while virtual circuit technologies such as ATM and Frame Relay provide more structured packet forwarding for applications requiring greater predictability.

Lecture reference: Section 2.1 – Packet Switching: Datagram and Virtual Circuit Services, Slides 7–8.

Section B – OSI Reference Model and Protocol Architecture

Question 3. OSI Model Application to Industrial Network Troubleshooting

A PLC cannot communicate with an HMI over an Ethernet network. The physical cable is connected, but the HMI displays a communication failure. A maintenance engineer decides to troubleshoot the problem systematically using the OSI reference model. Explain how the engineer can use the OSI layers to identify possible causes of the communication failure. Identify the major functions of the relevant layers and explain why the OSI model is valuable for troubleshooting.

Model Answer:

1. At the Physical Layer, the engineer checks Ethernet cabling, connectors, link status, and the electrical or optical signals required for communication.
2. At the Data Link Layer, the engineer investigates Ethernet frames, MAC addressing, switch operation, and possible frame errors.
3. At the Network Layer, the engineer verifies IP addressing, subnet configuration, and routing requirements between the PLC and HMI.
4. At the Transport and Application Layers, the engineer checks communication ports, protocol compatibility, application settings, and whether the required service is available.
5. The OSI model provides a structured troubleshooting framework that separates communication functions into manageable layers, helping the engineer isolate the source of the problem.

Lecture reference: Section 2.2 – OSI Reference Model, Slides 14–22.

Question 4. OSI Model vs. TCP/IP Model

An engineering team is designing a communication network using Ethernet switches, IP routing, TCP communication, and industrial application software. Some engineers describe the network using the seven-layer OSI model, while others use the TCP/IP model. Compare the OSI and TCP/IP models, explain how their layers correspond, and discuss why both models remain useful in modern industrial networking.

Model Answer:

1. The OSI model is a seven-layer conceptual reference model that separates networking functions to support communication design, understanding, and troubleshooting.
2. The TCP/IP model represents the practical protocol architecture used in modern computer networks and the Internet.
3. OSI Application, Presentation, and Session Layers correspond generally to the TCP/IP Application Layer.
4. The Transport and Network Layers correspond to the TCP/IP Transport and Internet Layers, while OSI Data Link and Physical functions are grouped into the TCP/IP Network Access Layer.
5. The OSI model is useful for systematically analyzing network problems, while TCP/IP describes the protocols used to implement communication in real industrial Ethernet systems.

Lecture reference: Section 2.2 – OSI vs. TCP/IP, Slides 17–19.

Question 5. IEEE 802 Standards and Data Link Sublayers

An industrial Ethernet communication system must support several different network protocols while ensuring that devices can access the communication medium and correctly address Ethernet frames. Explain how IEEE 802 standards extend the OSI reference model, distinguish the responsibilities of the Logical Link Control (LLC) and Media Access Control (MAC) sublayers, and describe their importance in LAN communication.

Model Answer:

1. IEEE 802 standards define networking rules primarily associated with the Physical and Data Link Layers of LAN and MAN technologies.
2. The Data Link Layer is divided into the LLC and MAC sublayers to separate protocol interfacing from local medium access functions.
3. LLC provides an interface between the Data Link and Network Layers and helps identify which upper-layer protocol should receive the transmitted information.
4. MAC controls access to the communication medium and handles local hardware addressing through MAC addresses.
5. Together, these functions enable devices to share the network medium, identify local destinations, and exchange properly structured frames.

Lecture reference: Section 2.1 – IEEE 802 Extensions to the OSI Reference Model, Slides 9–10.

Question 6. OSI Layer Independence and Network Upgrading

A factory operates an industrial Ethernet network at 100 Mbps and plans to upgrade its communication infrastructure to 1 Gbps. The production management team is concerned that this upgrade may require major modifications to PLC applications and supervisory software. Explain how the layered structure of the OSI model supports network upgrades and analyze which networking components and functions are most directly affected by this change.

Model Answer:

1. The OSI model separates communication functions into layers so that changes in one layer do not necessarily require changes in every other layer.
2. Increasing Ethernet speed from 100 Mbps to 1 Gbps primarily affects the Physical and Data Link Layers.
3. Network interface cards, Ethernet switches, cabling suitability, and supported communication speeds must be evaluated during the upgrade.
4. Higher-layer functions such as IP addressing, transport protocols, and application communication can generally remain logically unchanged when compatible networking equipment is used.
5. This independence simplifies network modernization by allowing engineers to improve communication performance without redesigning the complete industrial application.

Lecture reference: Section 2.2 – Role of a Reference Model, Slide 16.

Section C – Transport, Network, and Data Link Functions

Question 7. TCP vs. UDP for Industrial Data Communication

A manufacturing plant uses one communication service to transfer important production records and another service to transmit frequent process measurements. The engineering team must choose between TCP and UDP for these communication functions. Compare TCP and UDP in terms of connection establishment, reliability, communication overhead, packet ordering, and retransmission. Explain how these characteristics influence industrial communication design.

Model Answer:

1. TCP is a connection-oriented transport protocol that provides reliable communication using acknowledgements, sequence numbers, and retransmission.
2. TCP ensures that application data is delivered in the correct order and uses flow control to manage communication between endpoints.
3. UDP is connectionless and has lower protocol overhead because it does not provide TCP-style sequencing, retransmission, or guaranteed delivery.
4. TCP is suitable for transferring production records where reliable and ordered data delivery is important.
5. UDP may be suitable for frequent measurements when lower overhead is desirable and the application can tolerate or manage lost data.

Lecture reference: Section 2.2 – Transport Layer, Slides 30–31.

Question 8. Transport Layer Multiplexing and Port Numbers

An industrial computer simultaneously runs an HMI application, a web-based monitoring service, and an engineering application. All three applications communicate through the same network interface. Explain how the Transport Layer supports simultaneous communication between multiple applications and discuss the importance of port numbers, segmentation, and multiplexing/demultiplexing.

Model Answer:

1. The Transport Layer provides end-to-end communication services between applications operating on different network devices.
2. It uses source and destination port numbers to identify the applications or services involved in communication.
3. Multiplexing allows several applications to transmit data through the same network interface using the available communication stack.
4. Demultiplexing directs incoming transport data to the correct application based on the relevant addressing and port information.
5. Segmentation divides larger application data into smaller transport units, making network transmission and data handling more manageable.

Lecture reference: Section 2.2 – Transport Layer, Slides 30–31.

Question 9. Network Layer Routing Between Industrial Subnets

A PLC is installed on one IP subnet while a production database server is connected to another subnet. Although both devices are connected to the facility's network infrastructure, they cannot communicate directly using local Ethernet delivery. Explain how the Network Layer enables communication between the two subnets and describe the responsibilities of IP addressing, the default gateway, routing tables, and packet forwarding.

Model Answer:

1. The Network Layer provides logical addressing and routing so devices located on different IP networks can exchange data.
2. The PLC uses the destination IP address to determine whether the destination is within its local subnet or requires routing.
3. When the destination belongs to another subnet, the PLC forwards the packet toward its configured default gateway.
4. The router examines the destination IP address, consults its routing table, and forwards the packet toward the destination network.
5. Correct IP addresses, subnet configurations, and gateway settings are essential for successful communication between the PLC and the database server.

Lecture reference: Section 2.2 – Network Layer, Slides 32–33.

Question 10. Ethernet Frame Structure and Error Detection

An industrial controller transmits production data over Ethernet. During transmission, electromagnetic interference causes some bits within the Ethernet frame to become corrupted. Describe the structure of an Ethernet frame, explain the functions of its header, payload, and trailer, and analyze how the Data Link Layer detects corrupted frames.

Model Answer:

1. An Ethernet frame contains a header, payload, and trailer that provide the information needed for local network transmission.
2. The header contains source and destination MAC addresses and information identifying the type of data carried.
3. The payload carries the Layer 3 packet containing the information intended for delivery.
4. The trailer contains a Frame Check Sequence (FCS), which uses a Cyclic Redundancy Check (CRC) to detect transmission errors.
5. The receiving device verifies the frame integrity, and frames with detected errors are normally discarded rather than delivered as valid data.

Lecture reference: Section 2.2 – Data Link Layer, Slides 34–35.

Section D – Media Access Protocols

Question 11. FDMA, TDMA, and CDMA Comparison

A wireless communication network must support multiple industrial devices sharing a limited communication spectrum. The network designer is evaluating Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA). Compare the operating principles of these three channelization techniques and evaluate their differences in channel sharing, synchronization, bandwidth utilization, and implementation complexity.

Model Answer:

1. FDMA divides the available frequency spectrum into separate frequency channels, allowing multiple users to transmit simultaneously using different frequency bands.
2. TDMA divides communication time into repeating slots, allowing users to share the same frequency while transmitting at different times.
3. CDMA allows multiple users to transmit simultaneously on the same frequency by assigning unique codes that receivers use to separate signals.
4. FDMA can waste assigned bandwidth when users are inactive, while TDMA requires accurate timing and synchronization.
5. CDMA supports efficient spectrum sharing but requires more sophisticated signal processing and careful management of differences in received signal power.

Lecture reference: Section 2.3 – Channelization, Slides 39–42.

Question 12. Static vs. Dynamic Channel Allocation

An industrial wireless system initially supports a fixed number of sensors that transmit continuously. The facility later expands to include mobile equipment and sensors that communicate only when necessary. Compare static and dynamic channel allocation and explain why the most efficient allocation method may change as the communication requirements of the system evolve.

Model Answer:

1. Static channel allocation permanently assigns a communication channel to a particular user or device.
2. Static allocation is suitable when the number of devices is known and their communication requirements remain relatively constant.
3. Dynamic channel allocation assigns communication resources only when devices require them and releases those resources when communication ends.
4. Dynamic allocation improves channel utilization when many devices transmit intermittently or frequently enter and leave the network.
5. A fixed industrial installation may benefit from static allocation, while a changing or mobile communication environment generally benefits from dynamic allocation.

Lecture reference: Section 2.3 – Channelization, Slide 39.

Question 13. Polling, Reservation, and Token Passing in Controlled Access

An industrial network requires organized communication between multiple devices, with minimal possibility of transmission collisions. The network designer is considering polling, reservation, and token-passing access methods. Compare these controlled-access techniques, explain how each regulates transmission opportunities, and evaluate their advantages and limitations for industrial communication.

Model Answer:

1. Polling uses a central controller that asks devices in sequence or according to priority whether they need to transmit.
2. Reservation allows devices to request communication opportunities in advance and transmit during their assigned time slots.
3. Token passing uses a special control message that circulates among devices, permitting only the token holder to transmit.
4. These methods reduce or prevent simultaneous transmission conflicts and provide more organized access than uncontrolled random transmission.
5. Polling may suffer from controller bottlenecks, reservation introduces scheduling overhead, and token passing requires mechanisms to manage or recover lost tokens.

Lecture reference: Section 2.3 – Controlled Access, Slides 43–45.

Question 14. CSMA/CD and Ethernet Collision Handling

A legacy manufacturing system connects several controllers through a shared Ethernet hub operating in half-duplex mode. During periods of heavy network activity, the system experiences reduced communication throughput caused by collisions. Explain how CSMA/CD operates, describe how collisions are detected and resolved, and analyze why modern switched full-duplex Ethernet provides better communication performance.

Model Answer:

1. CSMA/CD allows a device to listen to the shared communication medium before beginning transmission.
2. When the medium appears idle, the device begins transmitting while monitoring for possible collisions.
3. If two devices transmit simultaneously and a collision occurs, the affected devices stop transmitting.
4. The devices wait for randomly selected periods using a binary exponential backoff procedure before attempting retransmission.
5. Modern switched full-duplex Ethernet eliminates normal shared-medium collisions by providing dedicated communication links, making CSMA/CD generally unnecessary.

Lecture reference: Section 2.3 – Random Access: CSMA/CD, Slide 47.

Question 15. ALOHA vs. Slotted ALOHA Efficiency

A communication designer is studying early random-access methods to understand how collisions affect the utilization of a shared communication channel. Two methods are compared: Pure ALOHA and Slotted ALOHA. Explain the operating principles of Pure ALOHA and Slotted ALOHA, compare their theoretical maximum channel utilization, and analyze why controlling transmission timing improves communication efficiency.

Model Answer:

1. Pure ALOHA allows devices to transmit whenever they have data, without first reserving a communication opportunity.
2. When transmitted messages collide, the affected devices wait for random periods before attempting retransmission.
3. Pure ALOHA has a maximum theoretical channel utilization of approximately 18%, because overlapping transmissions can frequently cause collisions.
4. Slotted ALOHA restricts transmissions to defined time boundaries and increases maximum theoretical utilization to approximately 37%.
5. Synchronizing transmission opportunities reduces the vulnerable period for collisions, improving channel efficiency compared with unrestricted random transmission.

Lecture reference: Section 2.3 – Random Access: ALOHA, Slide 46.

Section E – Encapsulation, Network Devices, and Communication Media

Question 16. Encapsulation and De-Encapsulation in Industrial Ethernet

A PLC transmits a production status message to a monitoring computer using TCP/IP over Ethernet. The message must pass through the communication protocol stack before transmission and must be reconstructed when it reaches the destination. Explain the encapsulation and de-encapsulation processes and describe the information added or processed at the Transport, Network, Data Link, and Physical Layers.

Model Answer:

1. The Application Layer creates the production message, which is passed downward through the communication protocol stack.
2. The Transport Layer adds information such as source and destination ports and, when TCP is used, control information supporting sequencing and reliable delivery.
3. The Network Layer adds source and destination IP addresses, creating a packet that can be forwarded through IP networks.
4. The Data Link Layer creates an Ethernet frame by adding MAC addressing and error-detection information, while the Physical Layer converts the frame into transmitted signals.
5. At the receiving computer, the process occurs in reverse as each layer processes its corresponding information until the original production message reaches the monitoring application.

Lecture reference: Section 2.4 – Encapsulation Process, Slides 49–57.

Question 17. Network Interface Card (NIC) Operation and Frame Filtering

An industrial computer receives Ethernet traffic from several PLCs and other devices through a network switch. Its network interface card must determine which received frames should be processed and forwarded to the operating system. Explain how a NIC participates in Ethernet communication, describe the role of MAC addressing in frame acceptance, and discuss the purpose of unicast, broadcast, multicast, and promiscuous-mode operation.

Model Answer:

1. A NIC provides the hardware interface between a computer and the network medium and performs functions associated with the Physical and Data Link Layers.
2. During reception, it converts incoming physical signals into bits, reconstructs Ethernet frames, and checks frame integrity and destination addressing.
3. Unicast frames are addressed to a particular network interface, while broadcast frames are intended for all devices within the local broadcast domain.
4. Multicast frames are addressed to selected groups of devices rather than one individual device or all devices.
5. Promiscuous mode allows a NIC to receive frames beyond those normally addressed to it, supporting packet capture and troubleshooting with tools such as Wireshark.

Lecture reference: Section 2.5 – Network Interface Card, Slides 58–61.

Question 18. Client-Server Communication in Industrial Automation

A manufacturing facility uses an HMI to monitor a PLC, an engineering workstation to retrieve configuration information, and a database server to store production records. These devices exchange information using client-server communication. Explain the client-server communication model, identify the roles that different industrial devices may perform, and discuss why a device can act as both a client and a server.

Model Answer:

1. A client is a device or software application that requests data, resources, or services from another system.
2. A server provides the requested service by receiving the client's request, processing it, and sending an appropriate response.
3. An HMI may act as a client when requesting process measurements from a PLC that provides the requested information.
4. An engineering workstation may act as a client when accessing a PLC or server, while a database server provides stored production data to authorized applications.
5. A device can function as both a client and a server depending on the communication activity, because these roles are defined by the services requested or provided rather than by hardware type alone.

Lecture reference: Section 2.6 – Clients and Servers, Slides 62–64.

Question 19. Selecting Network Media for an Industrial Plant

An automated factory requires three different communication connections: short-distance connections between PLCs and local switches, a long-distance connection between production buildings, and mobile connectivity for wireless engineering devices. The plant also contains equipment that generates electromagnetic interference. Compare twisted-pair copper, fiber-optic, and wireless communication media and recommend the most appropriate medium for each application. Justify your decisions based on distance, reliability, interference immunity, and flexibility.

Model Answer:

1. Twisted-pair copper Ethernet is suitable for short connections between PLCs and switches because it is relatively inexpensive, easy to install, and commonly supports distances up to approximately 100 meters.
2. Fiber-optic communication is appropriate for links between production buildings because it supports longer distances and higher transmission capacities.
3. Fiber optics are also preferable in electrically noisy environments because optical signals are resistant to electromagnetic interference.
4. Wireless LAN technology is suitable for mobile engineering devices because it provides flexible connectivity without requiring physical cables to each device.
5. A practical industrial design uses copper for suitable local equipment connections, fiber for long-distance or high-EMI links, and Wi-Fi where mobility is required, while considering wireless interference and security.

Lecture reference: Section 2.7 – Network Media, Slides 65–70.

Question 20. Ethernet Standard Selection and Network Modernization

A manufacturing plant uses older PLCs and HMIs connected through 100BASE-TX Ethernet. The plant plans to introduce industrial computers, production servers, and machine-vision applications that generate significantly larger amounts of data. Compare 100BASE-TX, 1000BASE-T, 10GBASE-T, and fiber-based Ethernet technologies. Recommend a practical network upgrade strategy that considers bandwidth, cabling requirements, equipment compatibility, and industrial operating conditions.

Model Answer:

1. 100BASE-TX provides 100 Mbps communication over twisted-pair copper and remains common in older PLCs, HMIs, and industrial equipment.
2. 1000BASE-T provides 1 Gbps communication using all four twisted pairs of Cat 5e or better cabling and is suitable for general industrial LANs.
3. 10GBASE-T provides 10 Gbps communication over suitable high-performance copper cabling and supports higher-bandwidth applications such as servers and machine-vision systems.
4. Fiber-based Ethernet standards provide alternatives for high-speed communication over longer distances and in environments with significant electromagnetic interference.
5. A practical upgrade retains compatible 100 Mbps connections for legacy devices while introducing Gigabit Ethernet for modern equipment and higher-capacity copper or fiber links for servers, backbones, and data-intensive applications.

Lecture reference: Section 2.8 – Ethernet Standards, Slides 71–74.